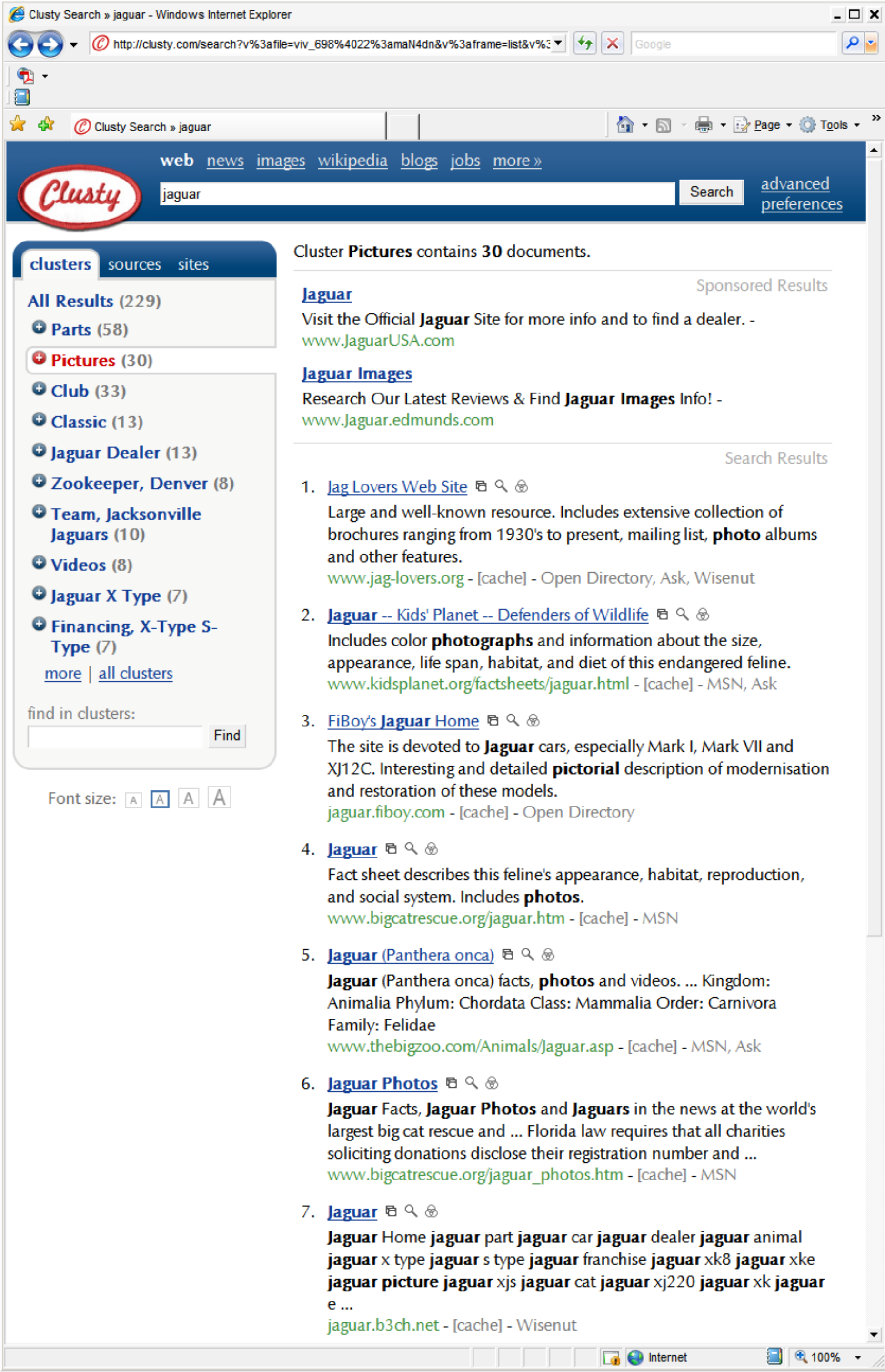
Clustering

Idea and Applications

- Clustering is the process of grouping a set of physical or abstract objects into classes of similar objects.
  - It is also called unsupervised learning.
  - It is a common and important task that finds many applications.
- Applications in Search engines:
  - Structuring search results
  - Suggesting related pages
  - Automatic directory construction/update
  - Finding near identical/duplicate pages

Improves recall
Allows disambiguation
Recovers missing details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Team, Jacksonville Jaguars (10)
Videos (8)
Jaguar X Type (7)
Financing, X-Type S-Type (7)
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Search Results
1. Jag Lovers Web Site
Large and well-known resource. Includes extensive collection of brochures ranging from 1930's to present, mailing list, photo albums and other features.
www.jag-lovers.org - [cache] - Open Directory, Ask, Wisenut
2. Jaguar -- Kids' Planet -- Defenders of Wildlife
Includes color photographs and information about the size, appearance, life span, habitat, and diet of this endangered feline.
www.kidsplanet.org/factsheets/jaguar.html - [cache] - MSN, Ask
3. FiBoy's Jaguar Home
The site is devoted to Jaguar cars, especially Mark I, Mark VII and XJ12C. Interesting and detailed pictorial description of modernisation and restoration of these models.
jaguar.fiboy.com - [cache] - Open Directory
4. Jaguar
Fact sheet describes this feline's appearance, habitat, reproduction, and social system. Includes photos.
www.bigcatrescue.org/jaguar.htm - [cache] - MSN
5. Jaguar (Panthera onca)
Jaguar (Panthera onca) facts, photos and videos. ... Kingdom: Animalia Phylum: Chordata Class: Mammalia Order: Carnivora Family: Felidae
www.thebigzoo.com/Animals/Jaguar.asp - [cache] - MSN, Ask
6. Jaguar Photos
Jaguar Facts, Jaguar Photos and Jaguars in the news at the world's largest big cat rescue and ... Florida law requires that all charities soliciting donations disclose their registration number and ...
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7. Jaguar
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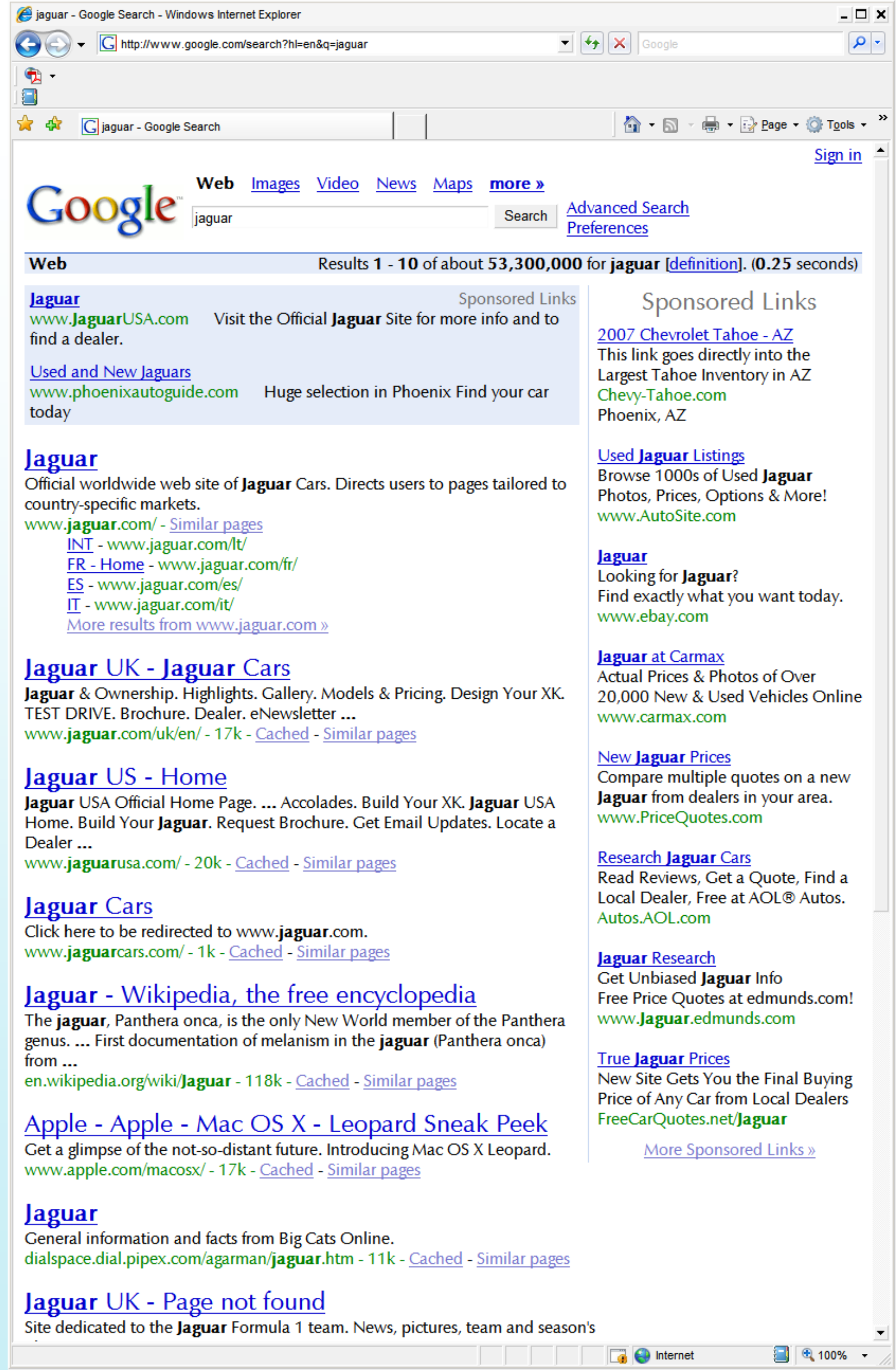
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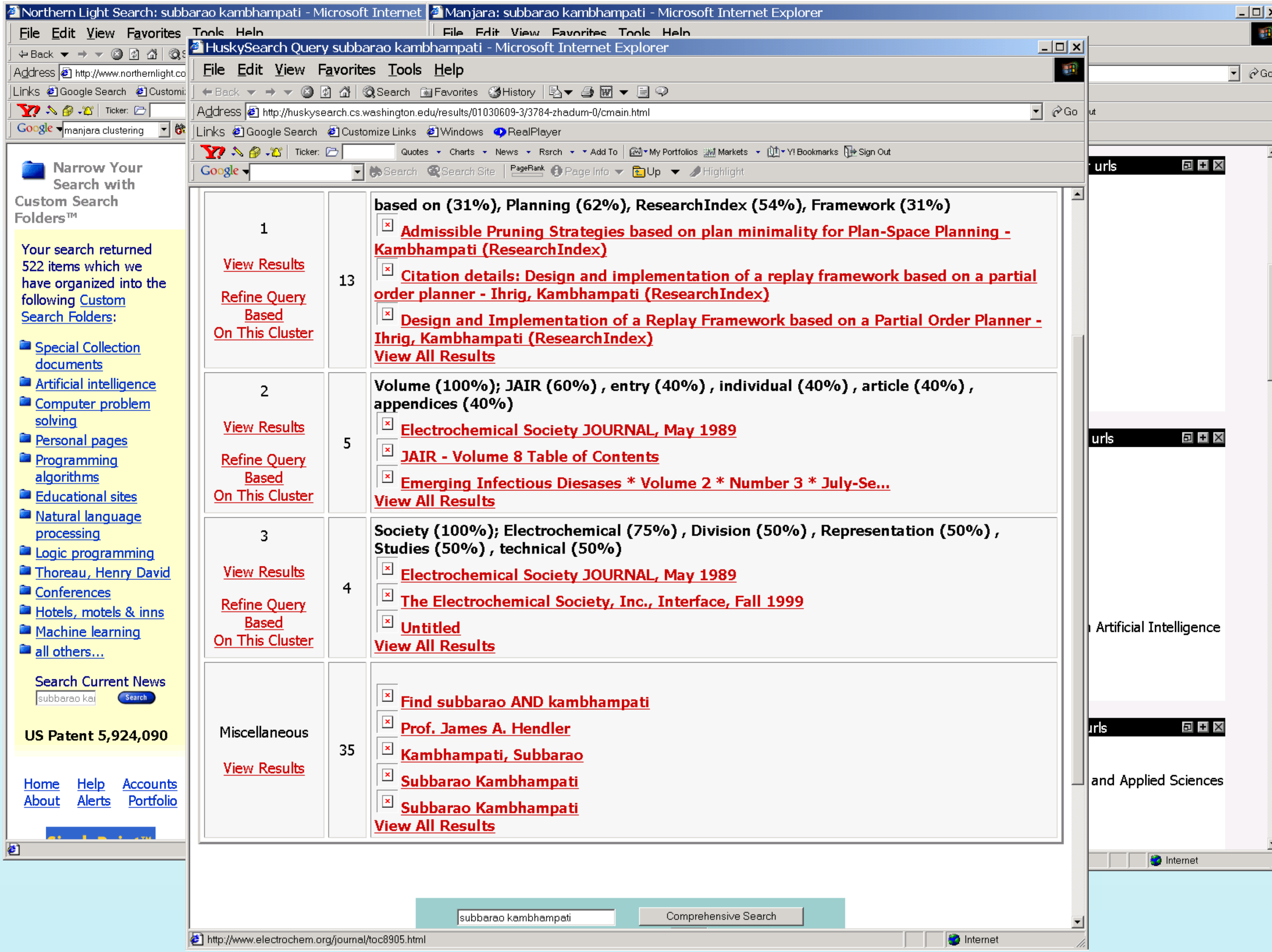
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based on (31%), Planning (62%), ResearchIndex (54%), Framework (31%)
Admissible Pruning Strategies based on plan minimality for Plan-Space Planning - Kambhampati (ResearchIndex)
Citation details: Design and implementation of a replay framework based on a partial order planner - Ihrig, Kambhampati (ResearchIndex)
Design and Implementation of a Replay Framework based on a Partial Order Planner - Ihrig, Kambhampati (ResearchIndex)
View All Results
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Volume (100%); JAIR (60%) , entry (40%) , individual (40%) , article (40%) , appendices (40%)
Electrochemical Society JOURNAL, May 1989
JAIR - Volume 8 Table of Contents
Emerging Infectious Diesases * Volume 2 * Number 3 * July-Se...
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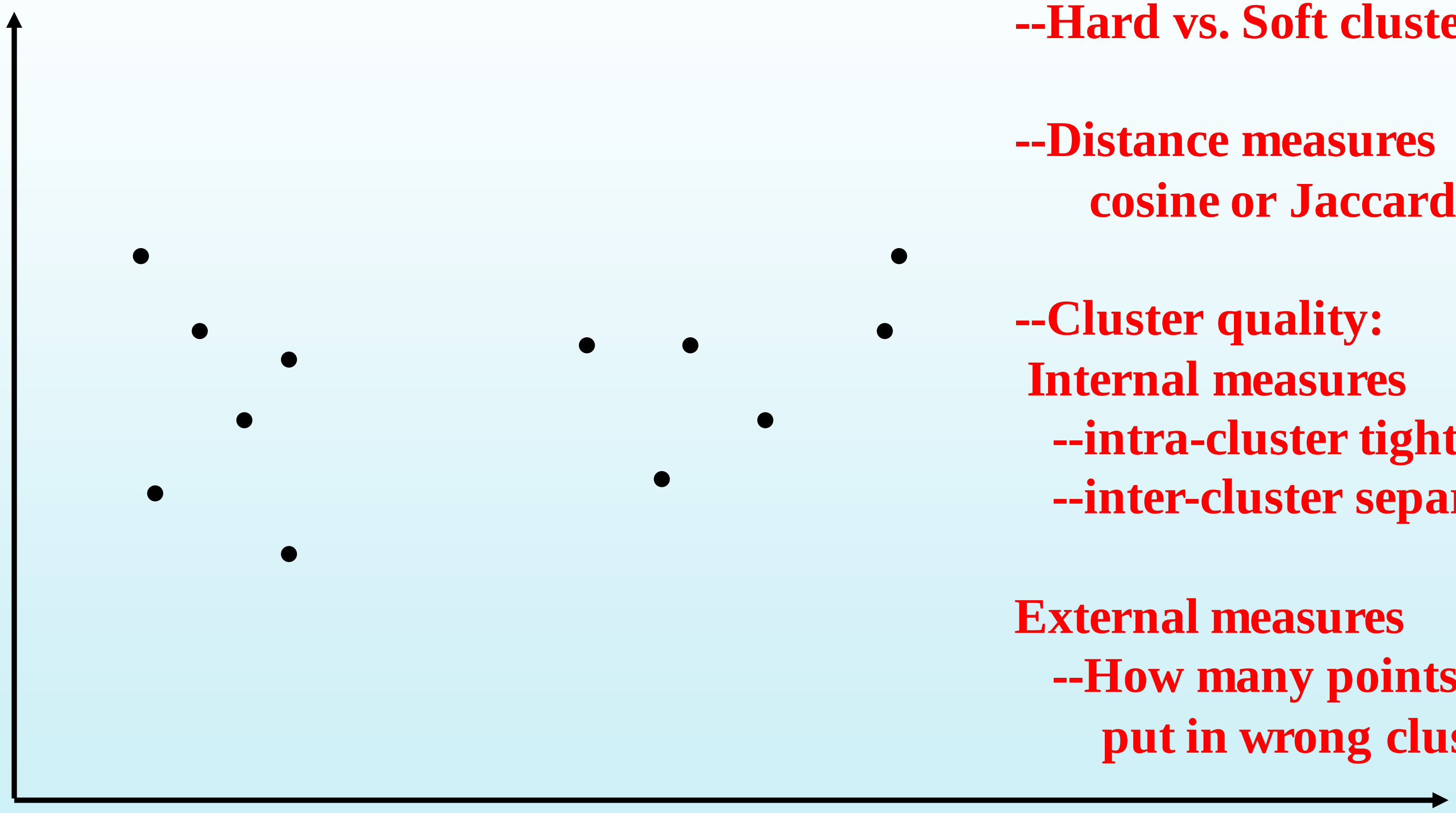
Clustering issues



--Hard vs. Soft clusters

--Distance measures
cosine or Jaccard or..

--Cluster quality:
Internal measures
--intra-cluster tightness
--inter-cluster separation

External measures
--How many points are put in wrong clusters.

[From Mooney]

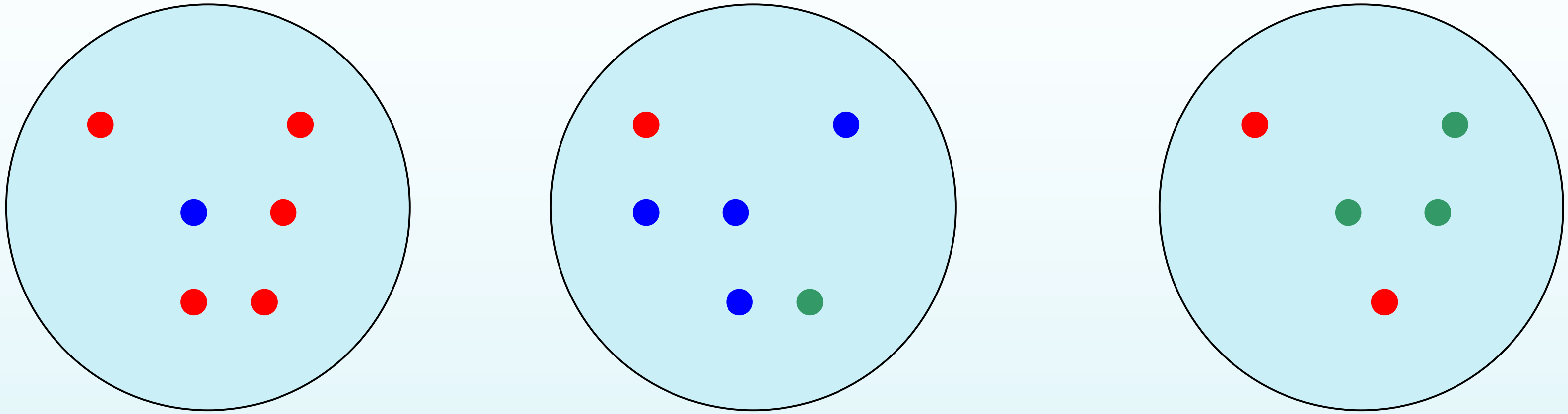
General issues in clustering

- Inputs/Specs
  - Are the clusters "hard" (each element in one cluster) or "Soft"
    - Hard Clustering=> partitioning
    - Soft Clustering=> subsets..
  - Do we know how many clusters we are supposed to look for?
    - Max # clusters?
    - Max possibilities of clusterings?
- What is a good cluster?
  - Are the clusters "close-knit"?
  - Do they have any connection to reality?
    - Sometimes we try to figure out reality by clustering…
- Importance of notion of distance
  - Sensitivity to outliers?

Cluster Evaluation

- “Clusters can be evaluated with “internal” as well as “external” measures
  - Internal measures are related to the inter/intra cluster distance
    - A good clustering is one where
      - » (Intra-cluster distance) the sum of distances between objects in the same cluster are minimized,
      - » (Inter-cluster distance) while the distances between different clusters are maximized
      - » Objective to minimize: F(Intra,Inter)
  - External measures are related to how representative are the current clusters to “true” classes. Measured in terms of purity, entropy or F-measure

Purity example



Cluster I Cluster II Cluster III

Cluster I: Purity = 1/6 (max(5, 1, 0)) = 5/6

Cluster II: Purity = 1/6 (max(1, 4, 1)) = 4/6

Cluster III: Purity = 1/5 (max(2, 0, 3)) = 3/5

Overall Purity = weighted purity

Rand-Index: Precision/Recall based

Number of points	Same Cluster in clustering	Different Clusters in clustering
Same class in ground truth	A	C
Different classes in ground truth	B	D

$$RI = \frac{A+D}{A+B+C+D}$$

$$P = \frac{A}{A+B}$$

$$R = \frac{A}{A+C}$$

Unsupervised?

- Clustering is normally seen as an instance of unsupervised learning algorithm
  - So how can you have external measures of cluster validity?
  - The truth is that you have a continuum between unsupervised vs. supervised
    - Answer: Think of “no teacher being there” vs. “lazy teacher” who checks your work once in a while.
    - Examples:
      - Fully unsupervised (no teacher)
      - Teacher tells you how many clusters are there
      - Teacher tells you that certain pairs of points will fall or will not fill in the same cluster
      - Teacher may occasionally evaluate the goodness of your clusters (external measures of validity)

(Text Clustering)
When & From What

- Clustering can be done at:
  - Indexing time
  - At query time
    - Applied to documents
    - Applied to snippets

Clustering can be based on:

URL source

Put pages from the same server together

Text Content

-Polysemy ("bat", "banks")

-Multiple aspects of a single topic

Links

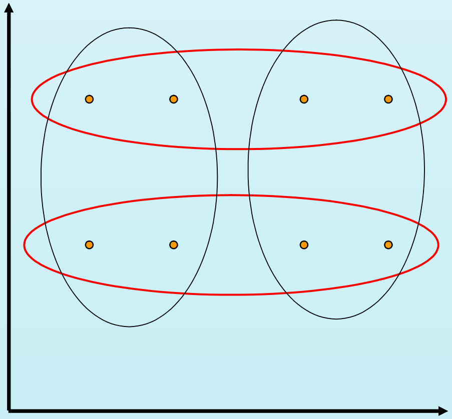
-Look at the connected components in the link graph (A/H analysis can do it)

-look at co-citation similarity (e.g. as in collab filtering)

Inter/Intra Cluster Distances

Intra-cluster distance/tightness

- (Sum/Min/Max/Avg) the (absolute/squared) distance between
  - All pairs of points in the cluster OR
  - Between the centroid and all points in the cluster OR
  - Between the "medoid" and all points in the cluster



Inter-cluster distance

Sum the (squared) distance between all pairs of clusters

Where distance between two clusters is defined as:

- distance between their centroids/medoids
- Distance between the closest pair of points belonging to the clusters (single link)
  - (Chain shaped clusters)
- Distance between farthest pair of points (complete link)
  - (Spherical clusters)

Entropy, F-Meas

Recall(i, j) = n_{ij} / n_i

Precision(i, j) = n_{ij} / n_j

$\sum_i p_{ij} \log(p_{ij}),$

Prob that a member of cluster j belongs to class i

Entropy of a clustering of

$$E_{CS} = \sum_{j=1}^{m} \frac{n_j * E_j}{n},$$

F(i, j) = (2 * Recall(i, j) * Precision(i, j)) / ((Precision(i, j) + Recall(i, j))

Cluster j class i

How hard is clustering?

- One idea is to consider all possible clusterings, and pick the one that has best inter and intra cluster distance properties
- Suppose we are given n points, and would like to cluster them into k-clusters
  - How many possible clusterings?

$$\frac{k^n}{k!}$$

- Too hard to do it brute force or optimally
- Solution: Iterative optimization algorithms
  - Start with a clustering, iteratively improve it (eg. K-means)

Classical clustering methods

- Partitioning methods
  - k-Means (and EM), k-Medoids
- Hierarchical methods
  - agglomerative, divisive, BIRCH
- Model-based clustering methods

K-means

- Works when we know k, the number of clusters we want to find
- Idea:
  - Randomly pick k points as the "centroids" of the k clusters
  - Loop:
    - For each point, put the point in the cluster to whose centroid it is closest
    - Recompute the cluster centroids
    - Repeat loop (until there is no change in clusters between two consecutive iterations.)

Iterative improvement of the objective function:

Sum of the squared distance from each point to the centroid of its cluster

(Notice that since K is fixed, maximizing tightness also maximizes inter-cluster distance)

Lower case

Convergence of *K*-Means

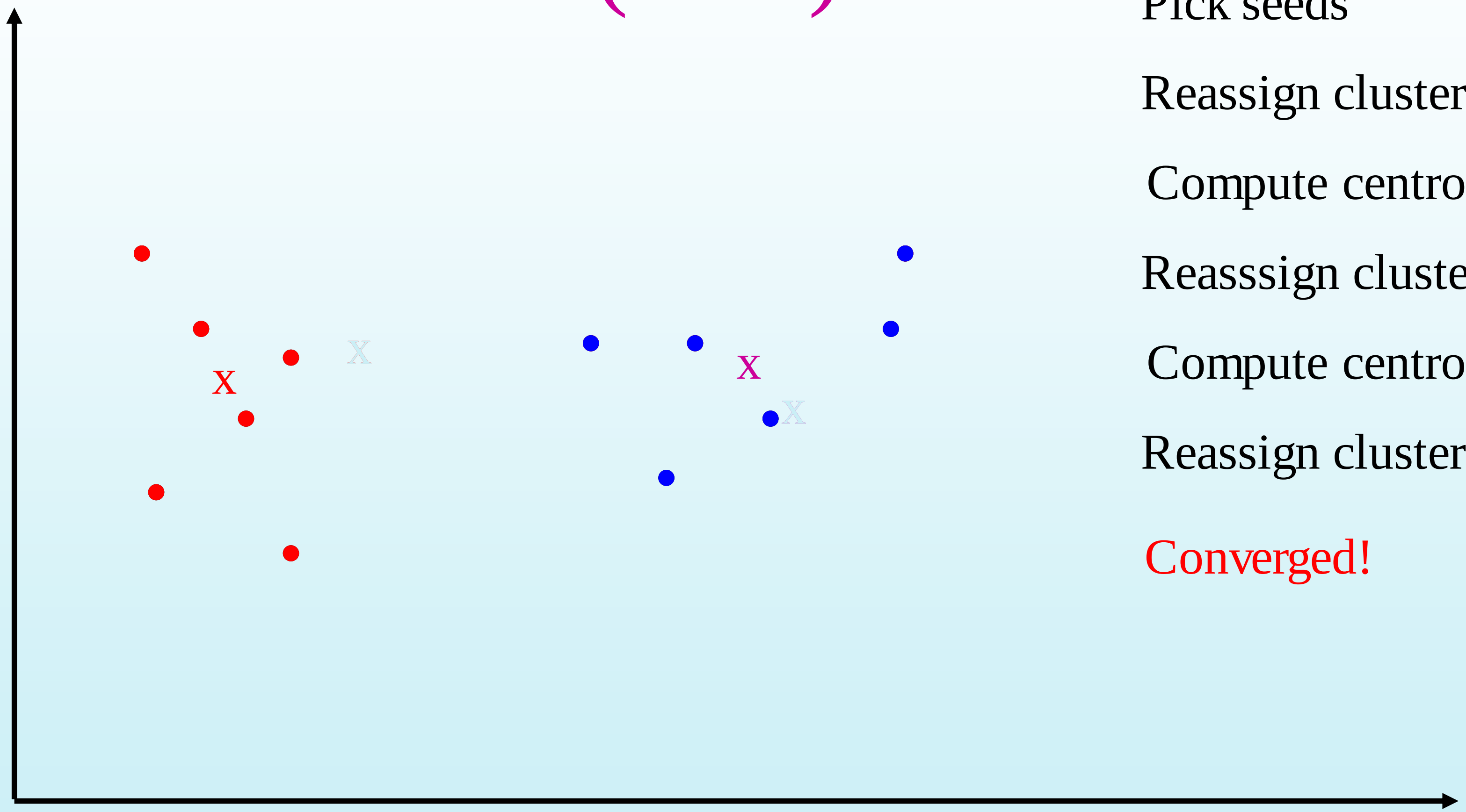
- Define goodness measure of cluster k as sum of squared distances from cluster centroid:
  - $G_k = \Sigma_i (d_i - c_k)^2$ (sum over all d_i in cluster k)
- $G = \Sigma_k G_k$
- Reassignment monotonically decreases G since each vector is assigned to the closest centroid.

K-means Example

- For simplicity, 1-dimension objects and k=2.
  - Numerical difference is used as the distance
- Objects: 1, 2, 5, 6,7
- K-means:
  - Randomly select 5 and 6 as centroids;
  - => Two clusters {1,2,5}and {6,7}; meanC1=8/3, meanC2=6.5
  - => {1,2}, {5,6,7}; meanC1=1.5, meanC2=6
  - => no change.
  - Aggregate dissimilarity
    - (sum of squares of distanceeach point of each cluster from its cluster center--(intra-cluster distance)
  - $= 0.5^2 + 0.5^2 + 1^2 + 0^2 + 1^2 = 2.5$

$|1-1.5|^2$

K Means Example (K=2)



Pick seeds

Reassign clusters

Compute centroids

Reasssign clusters

Compute centroids

Reassign clusters

Converged!

[From Mooney]

Example of K-means in operation

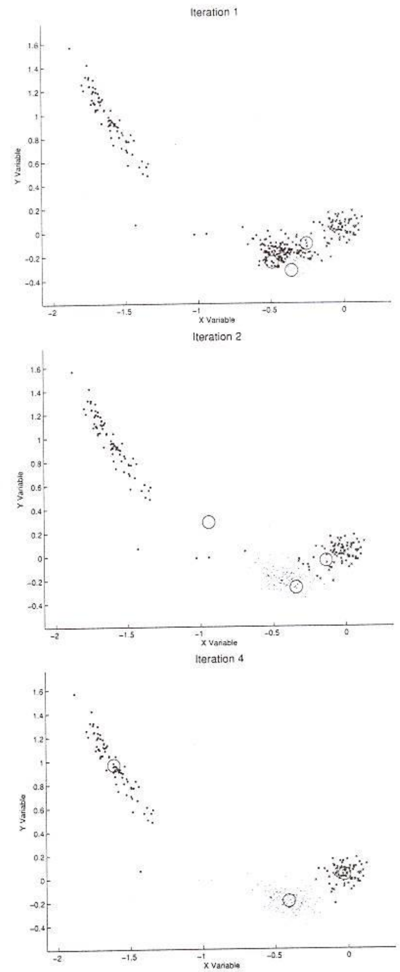


Figure 9.5 Example of running the K-means algorithm on the two-dimensional antenna data. The plots show the locations of the means of the clusters (large circles) at various iterations of the K-means algorithm, as well as the classification of the data points at each iteration according to the closest mean (dots, circles, and xs for each of the three clusters).

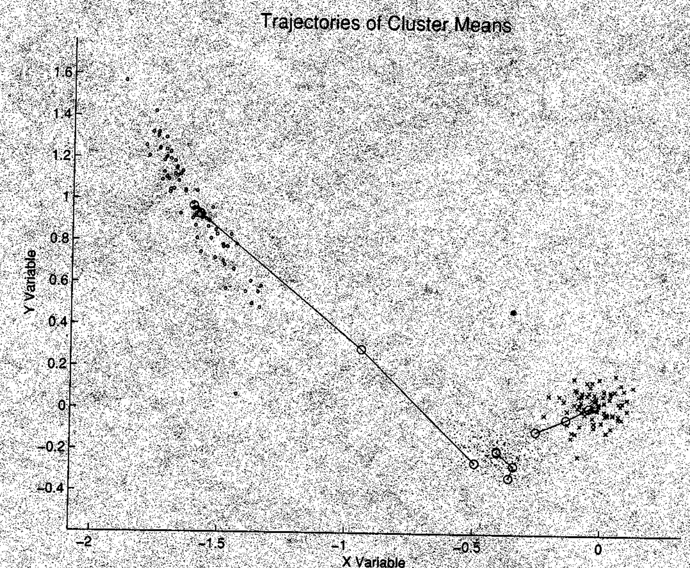


Figure 9.6 A summary of the trajectories of the three cluster means during the *K*-means iterations of figure 9.5.

[From Hand et. Al.]

Vector Quantization: K-means as Compression

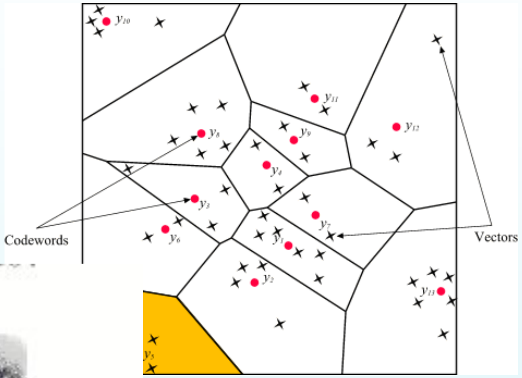


FIGURE 14.9. *Sir Ronald A. Fisher (1890-1962) was one of the founders of modern day statistics, to whom we owe maximum-likelihood, sufficiency, and many other fundamental concepts. The image on the left is a* 1024×1024 *grayscale image at 8 bits per pixel. The center image is the result of* 2×2 *block VQ, using 200 code vectors, with a compression rate of 1.9 bits/pixel. The right image uses only four code vectors, with a compression rate of 0.50 bits/pixel*

Problems with K-means

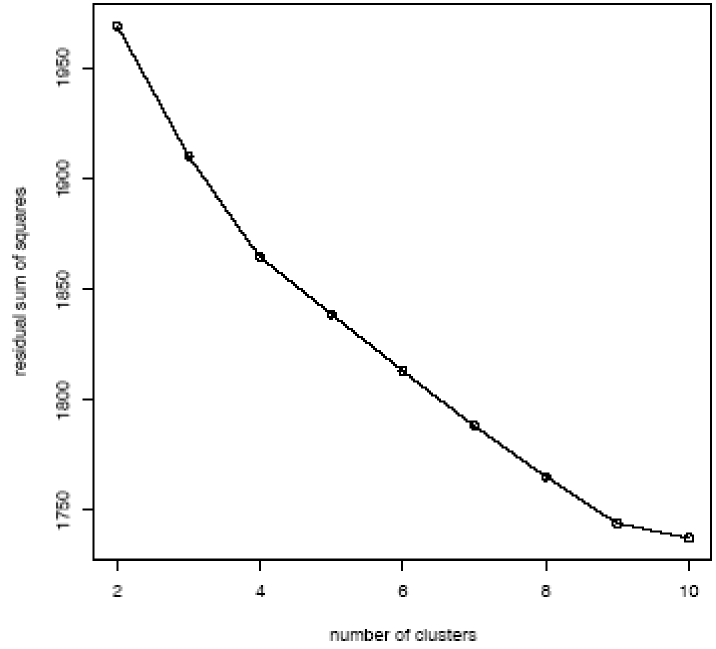
- Need to know k in advance
  - Could try out several k?
    - Cluster tightness increases with increasing K.
      - Look for a kink in the tightness vs. K curve
- Tends to go to local minima that are sensitive to the starting centroids
  - Try out multiple starting points
- Disjoint and exhaustive
  - Doesn't have a notion of "outliers"
    - Outlier problem can be handled by K-medoid or neighborhood-based algorithms
- Assumes clusters are spherical in vector space
  - Sensitive to coordinate changes, weighting etc.

Why not the minimum value?

Example showing sensitivity to seeds

In the above, if you start with B and E as centroids you converge to {A,B,C} and {D,E,F}
If you start with D and F you converge to {A,B,D,E} {C,F}

Looking for knees in the sum of intra-cluster dissimilarity



► **Figure 16.8** Minimal residual sum of squares (RSS) as a function of the number of clusters in k-means. In this clustering of 1203 Reuters-RCV1 documents, there are two points where the RSS curve flattens: at 4 clusters and at 9 clusters. The documents were selected from the categories *China*, *Germany*, *Russia* and *Sports*, so the $K = 4$ clustering is closest to the Reuters categorization.

Penalize lots of clusters

- For each cluster, we have a Cost C.
- Thus for a clustering with K clusters, the Total Cost is KC.
- Define the Value of a clustering to be =

  Total Benefit - Total Cost.
- Find the clustering of highest value, over all choices of K.
  - Total benefit increases with increasing K. But can stop when it doesn't increase by "much". The Cost term enforces this.

Time Complexity

- Assume computing distance between two instances is O(m) where m is the dimensionality of the vectors.
- Reassigning clusters: O(kn) distance computations, or O(knm).
- Computing centroids: Each instance vector gets added once to some centroid: O(nm).
- Assume these two steps are each done once for I iterations: O(Iknm).
- Linear in all relevant factors, assuming a fixed number of iterations,
  - more efficient than O(n^2) HAC (to come next)

Centroid Properties..

$$c = \frac{1}{|S|}\sum_{d \in S} d$$

$$cosine(d, c) = (d \bullet c) / \|d\| \, \|c\| = (d \bullet c) / \|c\|$$

$$cosine(c_1, c_2) = (c_1 \bullet c_2) / \|c_1\| \, \|c_2\|$$

$$d_1 \bullet c = \frac{1}{|S|}\sum_{d \in S} d_1 \bullet d = \frac{1}{|S|}\sum_{d \in S} cosine(d_1, d)$$

Similarity between a doc and the centroid is equal to avg similarity between that doc and every other doc

$$\frac{1}{|S|^2}\sum_{\substack{d \in S \\ d' \in S}} cosine(d', d) = \frac{1}{|S|}\sum_{d \in S} d \bullet \frac{1}{|S|}\sum_{d \in S} d = c \bullet c = \|c\|^2$$

Average similarity between all pairs of documents is equal to the square of centroid's magnitude.

Variations on K-means

- Recompute the centroid after every (or few) changes (rather than after all the points are re-assigned)
  - Improves convergence speed
- Starting centroids (seeds) change which local minima we converge to, as well as the rate of convergence
  - Use heuristics to pick good seeds
    - Can use another cheap clustering over random sample
  - Run K-means M times and pick the best clustering that results
    - Bisecting K-means takes this idea further…

Lowest aggregate Dissimilarity (intra-cluster distance)

Hybrid method 1

Bisecting K-means

Can pick the largest Cluster or the cluster With lowest average similarity

- For I=1 to k-1 do{
  - Pick a leaf cluster C to split
  - For J=1 to ITER do{
    - Use K-means to split C into two sub-clusters, C_1 and C_2
    - Choose the best of the above splits and make it permanent}

}

Divisive hierarchical clustering method uses K-means

Variations on K-means (contd)

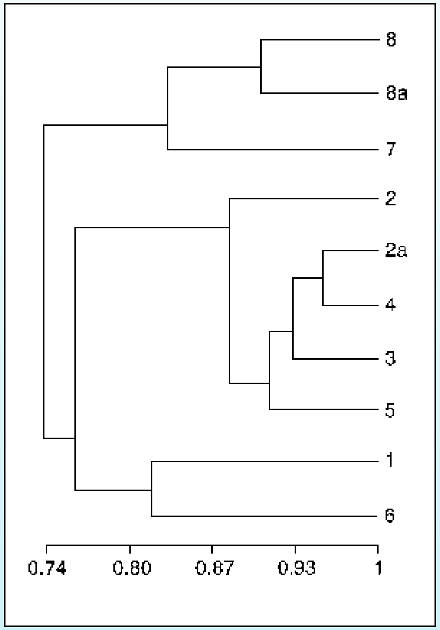
- Outlier problem
  - Use K-Medoids
    - Costly!
- Non-hard clusters
  - Use soft K-means
    - Let the membership of each data point in a cluster be proportional to its distance from that cluster center
    - Membership weight of elt e in cluster C is Exp(-b dist(e; center(C))
      - » Normalize the weight vector
  - Normal K-means takes the max of weights and assigns it to that cluster
    - » The cluster center re-computation step is based on the membership
  - We can instead let the cluster center computation be based on the all points, weighted by their membership weight

Semi-supervised variations of K-means

- Often we know partial knowledge about the clusters
  - E.g. We may know that the text docs are in two clusters—one related to finance and the other to CS.
  - Moreover, we may know that certain specific docs are CS and certain others are finance
  - How can we use it?
- Partial knowledge can be used to set the initial seeds of the K-means

Hierarchical Clustering Techniques

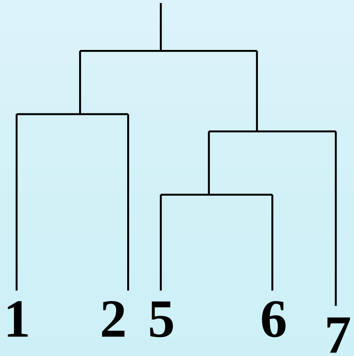
- Generate a nested (multi-resolution) sequence of clusters
- Two types of algorithms
  - Divisive
    - Start with one cluster and recursively subdivide
    - Bisecting K-means is an example!
  - Agglomerative (HAC)
    - Start with data points as single point clusters, and recursively merge the closest clusters



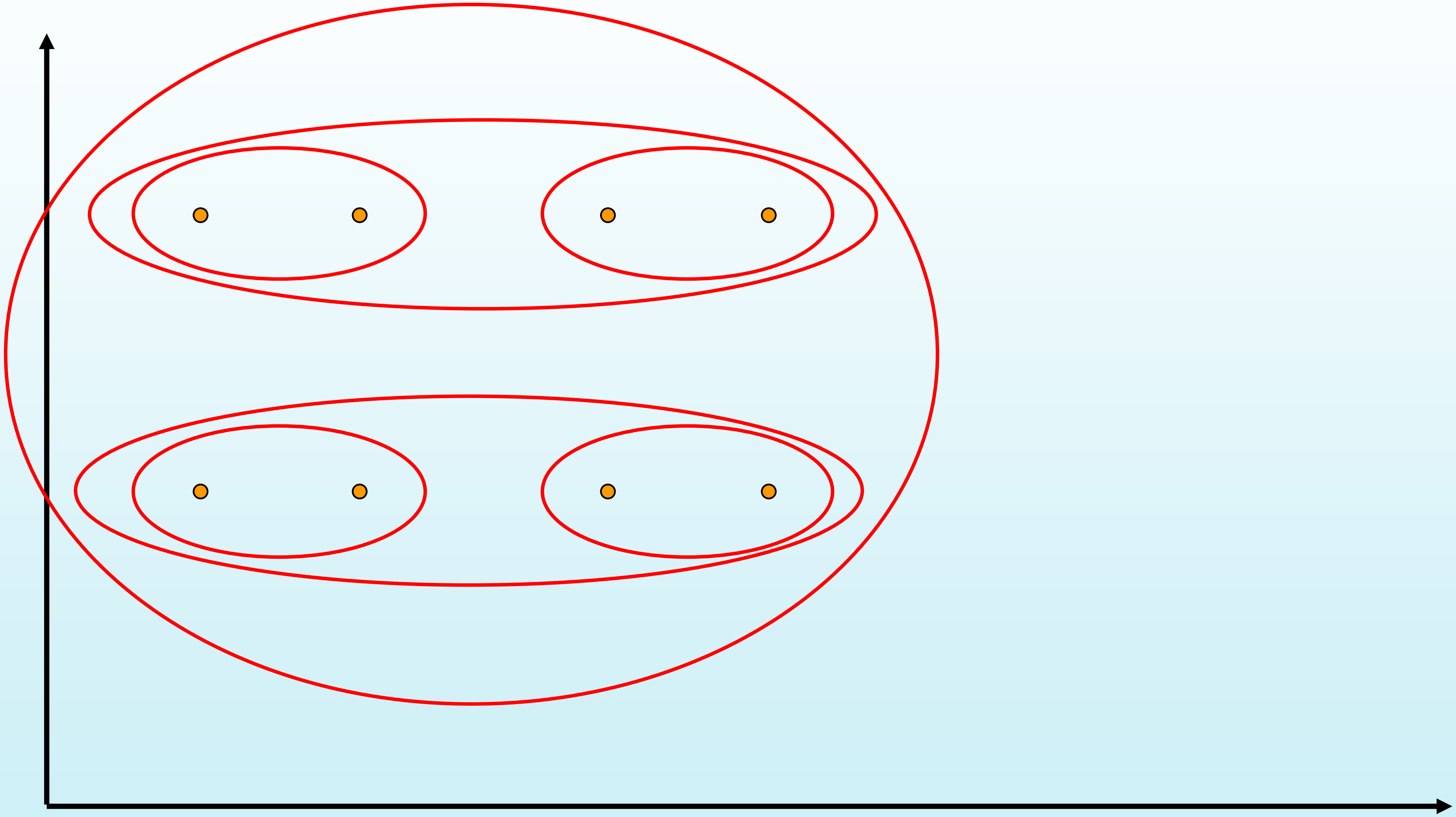
“Dendogram”

Hierarchical Agglomerative Clustering Example

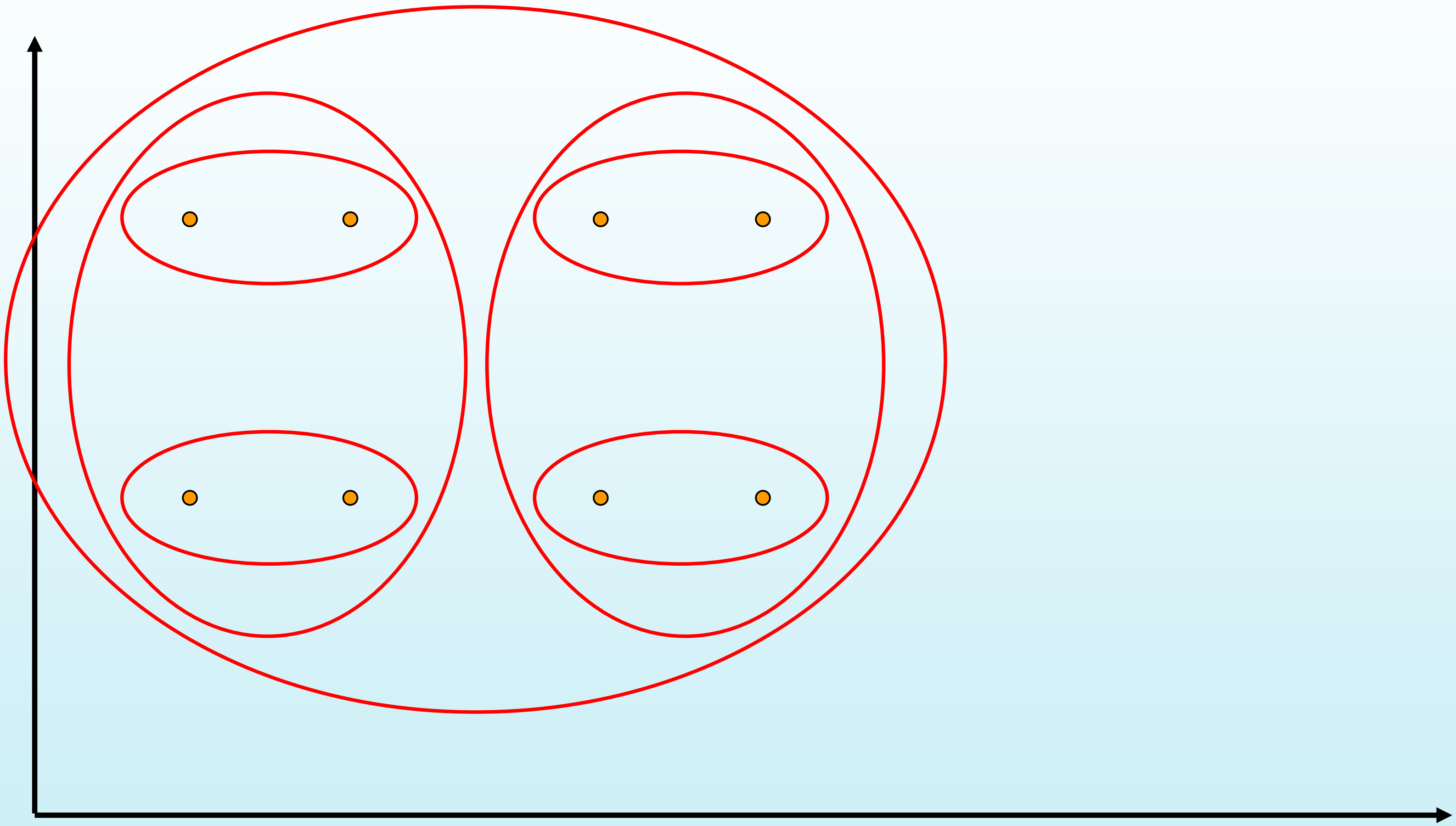
- {Put every point in a cluster by itself.
  For I=1 to N-1 do{
  let C_1 and C_2 be the most mergeable pair of clusters
  →(defined as the two closest clusters)
  Create $C_{1,2}$ as parent of C_1 and C_2}
- Example: For simplicity, we still use 1-dimensional objects.
  - Numerical difference is used as the distance
- Objects: 1, 2, 5, 6,7
- agglomerative clustering:
  - find two closest objects and merge;
  - => {1,2}, so we have now {1.5,5, 6,7};
  - => {1,2}, {5,6}, so {1.5, 5.5,7};
  - => {1,2}, {{5,6},7}.



Single Link Example

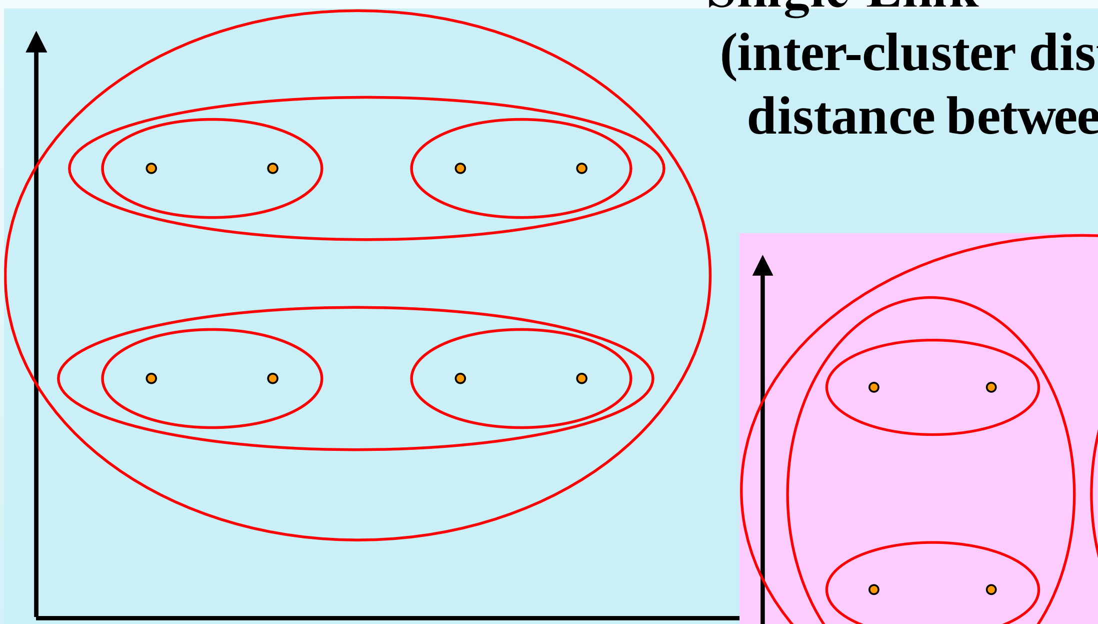


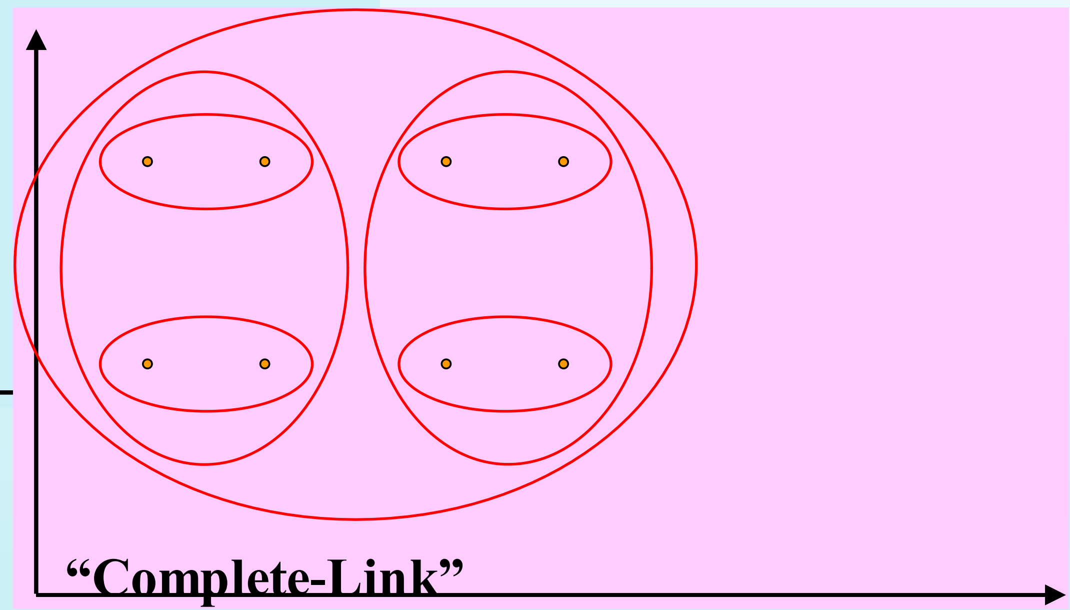
Complete Link Example



Impact of cluster distance measures

"Single-Link"
(inter-cluster distance=
distance between closest pair of points)





"Complete-Link"
(inter-cluster distance=
distance between farthest pair of points)

[From Mooney]

Group-average Similarity based clustering

- Instead of single or complete link, we can consider cluster distance in terms of average distance of all pairs of points from each cluster
- Problem: n*m similarity computations
- Thankfully, this is much easier with cosine similarity…

$$\frac{1}{|c1||c2|}\sum_{di\in C1}\sum_{dj\in C2} di \bullet dj = \frac{1}{|c1|}\sum_{di\in C1} di \bullet \frac{1}{|c2|}\sum_{dj\in C2} d2$$

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$$\frac{1}{|S|^2}\sum_{\substack{d\in S\\ d'\in S}} cosine(d',d) = \frac{1}{|S|}\sum_{d\in S} d \bullet \frac{1}{|S|}\sum_{d\in S} d = c \bullet c = \|c\|^2$$

Average similarity between all pairs of documents is equal to the square of centroid's magnitude.

Properties of HAC

- Creates a complete binary tree ("Dendogram") of clusters
- Various ways to determine mergeability
  - "Single-link"—distance between closest neighbors
  - "Complete-link"—distance between farthest neighbors
  - "Group-average"—average distance between all pairs of neighbors
  - "Centroid distance"—distance between centroids is the most common measure
- Deterministic (modulo tie-breaking)
- Runs in $O(N^2)$ time
- People used to say this is better than K-means
    - But the Stenbach paper says K-means and bisecting K-means are actually better

Already discussed

Hybrid method 1

Bisecting K-means

Can pick the largest Cluster or the cluster With lowest average similarity

- For I=1 to k-1 do{
  - Pick a leaf cluster C to split
  - For J=1 to ITER do{
    - Use K-means to split C into two sub-clusters, C_1 and C_2
    - Choose the best of the above splits and make it permanent}

}

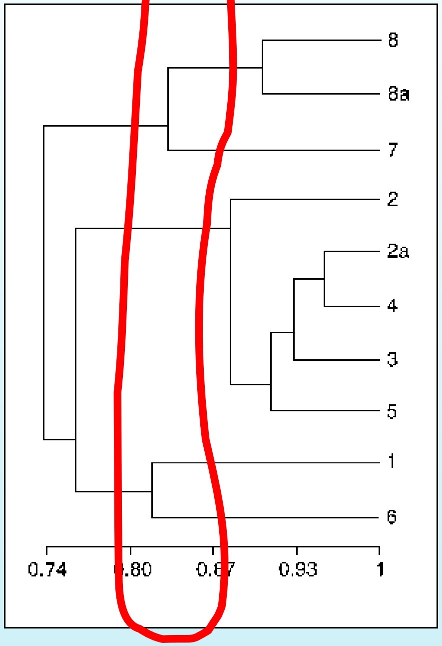
Divisive hierarchical clustering method uses K-means

Hybrid method 2

Buckshot Algorithm

- Combines HAC and K-Means clustering.
- First randomly take a sample of instances of size $\sqrt{n}$
- Run group-average HAC on this sample, which takes only O(n) time.
- Use the results of HAC as initial seeds for K-means.
- Overall algorithm is O(n) and avoids problems of bad seed selection.

Cut where You have k clusters



Uses HAC to bootstrap K-means

Text Clustering

- HAC and K-Means have been applied to text in a straightforward way.
- Typically use normalized, TF/IDF-weighted vectors and cosine similarity.
- Cluster Summaries are computed by using the words that have highest tf/icf value (i.c.f→Inverse cluster frequency)
- Optimize computations for sparse vectors.
- Applications:
  - During retrieval, add other documents in the same cluster as the initial retrieved documents to improve recall.
  - Clustering of results of retrieval to present more organized results to the user (à la Northernlight folders).
  - Automated production of hierarchical taxonomies of documents for browsing purposes (à la Yahoo & DMOZ).

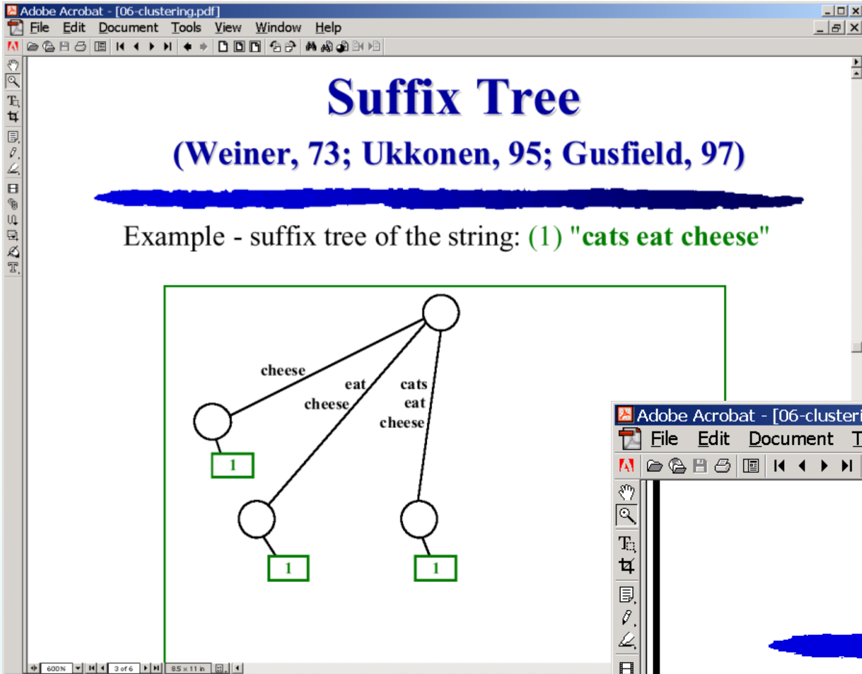
Which of these are the best for text?

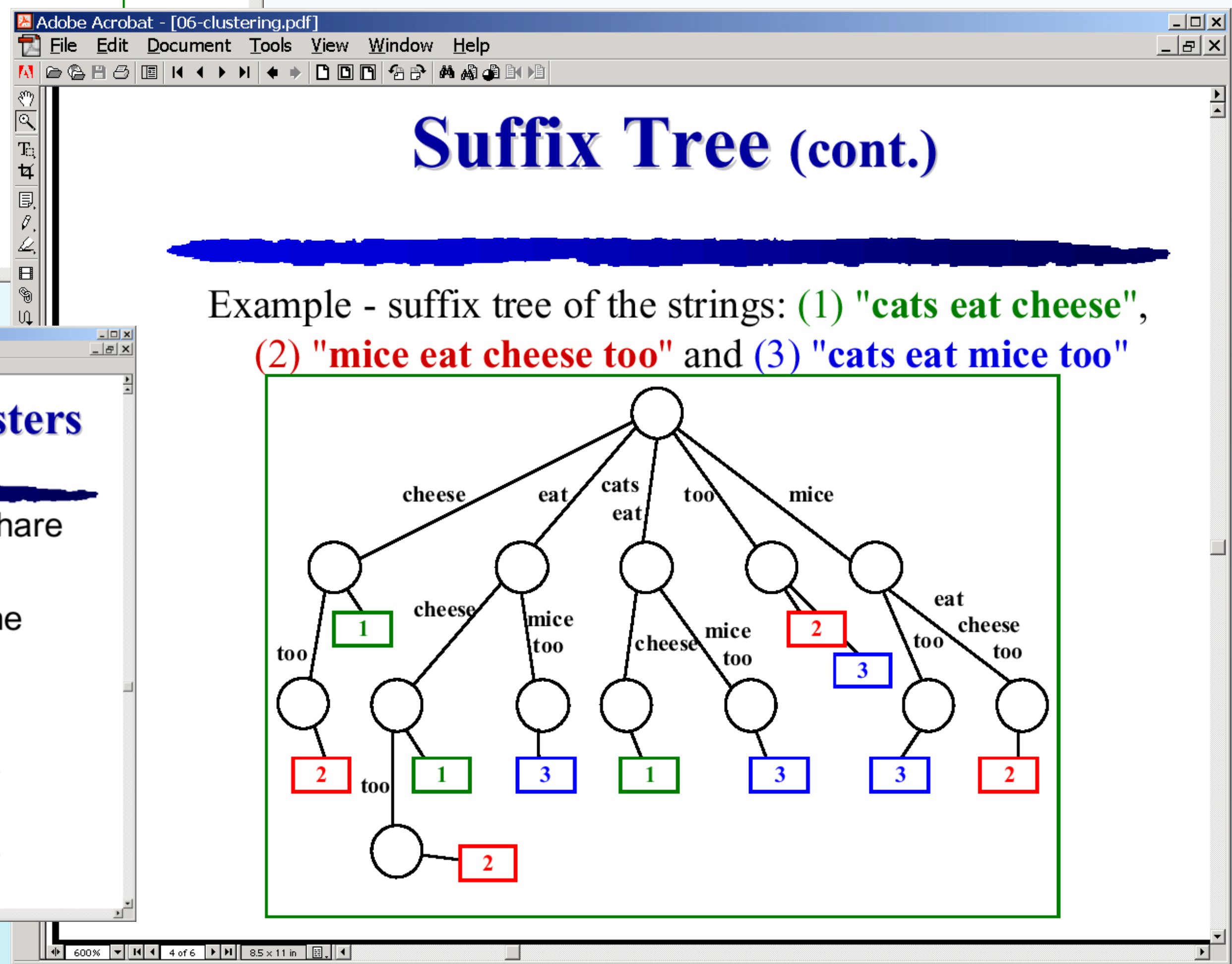
- Bisecting K-means and K-means seem to do better than Agglomerative Clustering techniques for Text document data [Steinbach et al]
  - "Better" is defined in terms of cluster quality
    - Quality measures:
      - Internal: Overall Similarity
      - External: Check how good the clusters are w.r.t. user defined notions of clusters

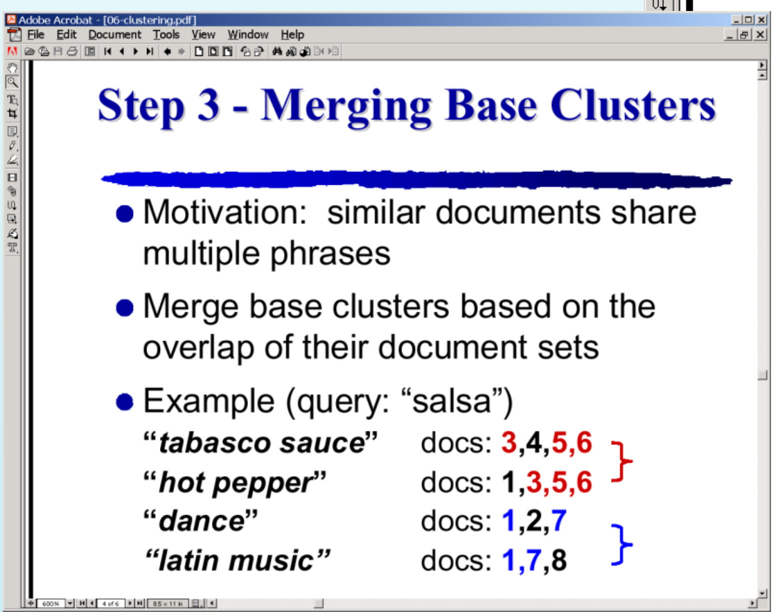
Challenges/Other Ideas

- High dimensionality
  - Most vectors in high-D spaces will be orthogonal
  - Do LSI analysis first, project data into the most important m-dimensions, and then do clustering
    - E.g. Manjara
- Phrase-analysis (a better distance and so a better clustering)
  - Sharing of phrases may be more indicative of similarity than sharing of words
    - (For full WEB, phrasal analysis was too costly, so we went with vector similarity. But for top 100 results of a query, it is possible to do phrasal analysis)
    - Suffix-tree analysis
    - Shingle analysis
- Using link-structure in clustering
  - A/H analysis based idea of connected components
  - Co-citation analysis
    - Sort of the idea used in Amazon's collaborative filtering
- Scalability
  - More important for "global" clustering
  - Can't do more than one pass; limited memory
  - See the paper
    - Scalable techniques for clustering the web
    - Locality sensitive hashing is used to make similar documents collide to same buckets

Phrase-analysis based similarity (using suffix trees)







Other (general clustering) challenges

- Dealing with noise (outliers)
  - "Neighborhood" methods
    - "An outlier is one that has less than δ points within ε distance" (δ, ε pre-specified thresholds)
    - Need efficient data structures for keeping track of neighborhood
      - R-trees
- Dealing with different types of attributes
  - Hard to define distance over categorical attributes